

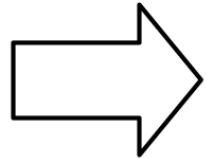
Reshaping data

Logistics and reminders

- HW 1 due tonight
- HW 2 released, due next Friday

Last time: `pivot_longer`

id	bp1	bp2
A	100	120
B	140	115
C	120	125



id	measurement	value
A	bp1	100
A	bp2	120
B	bp1	140
B	bp2	115
C	bp1	120
C	bp2	125

```
1 df |>
2   pivot_longer(
3     cols = bp1:bp2,
4     names_to = "measurement",
5     values_to = "value"
6   )
```

Why pivot?

```
# A tibble: 260 × 38
```

```
  country `1975` `1976` `1977` `1978` `1979` `1980` `1981` `1982`  
`1983` `1984`  
  <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>  
<dbl> <dbl>  
1 Afghan...    NA    NA    NA    NA    4.99    NA    NA    NA  
NA    NA  
2 Albania     NA    NA    NA    NA    NA     NA    NA    NA  
NA    NA  
3 Algeria     NA    NA    NA    NA    NA     NA    NA    NA  
NA    NA  
4 Andorra     NA    NA    NA    NA    NA     NA    NA    NA  
NA    NA
```

Challenge: a variable of interest (year) is contained in the column names!

Why pivot?

Literacy data in tidy form:

```
1 litF_long <- litF |>
2   pivot_longer(
3     cols = -country,
4     names_to = "year",
5     values_to = "literacy_rate",
6     values_drop_na = T
7   )
8
9 litF_long
```

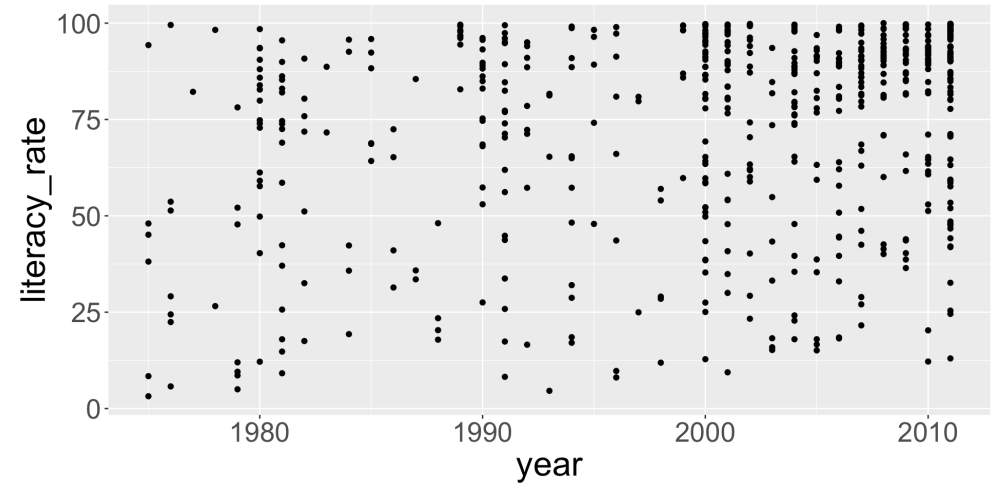
A tibble: 571 × 3

	country	year	literacy_rate
	<chr>	<chr>	<dbl>
1	Afghanistan	1979	4.99
2	Afghanistan	2011	13
3	Albania	2001	98.3
4	Albania	2008	94.7
5	Albania	2011	95.7
6	Algeria	1987	35.8

7	Algeria	2002	60.1
8	Algeria	2006	63.9
9	Angola	2001	54.2
10	Angola	2011	58.6

Why pivot?

	country	year	literacy_rate
1	Afghanistan	1979	4.987460
2	Afghanistan	2011	13.000000
3	Albania	2001	98.252274
4	Albania	2008	94.681814
5	Albania	2011	95.691480
6	Algeria	1987	35.839915
7	Algeria	2002	60.075082
8	Algeria	2006	63.918785
9	Angola	2001	54.194488
10	Angola	2011	58.608460
11	Anguilla	1984	95.714930
12	Antigua and Barbuda	2001	99.420000
13	Antigua and Barbuda	2011	99.420000
14	Argentina	1980	93.580894
15	Argentina	1991	96.041358
16	Argentina	2001	97.193411
17	.	.	.



```
1 litF_long |>
2   ggplot(aes(x = year,
3               y = literacy_rate)) +
4   geom_point()
```

Why pivot?

	country	year	literacy_rate
1	Afghanistan	1979	4.987460
2	Afghanistan	2011	13.000000
3	Albania	2001	98.252274
4	Albania	2008	94.681814
5	Albania	2011	95.691480
6	Algeria	1987	35.839915
7	Algeria	2002	60.075082
8	Algeria	2006	63.918785
9	Angola	2001	54.194488
10	Angola	2011	58.608460
11	Anguilla	1984	95.714930
12	Antigua and Barbuda	2001	99.420000
13	Antigua and Barbuda	2011	99.420000
14	Argentina	1980	93.580894
15	Argentina	1991	96.041358
16	Argentina	2001	97.193411
17	.	.	.

(Intercept)	year
-1323.2098674	0.6979597

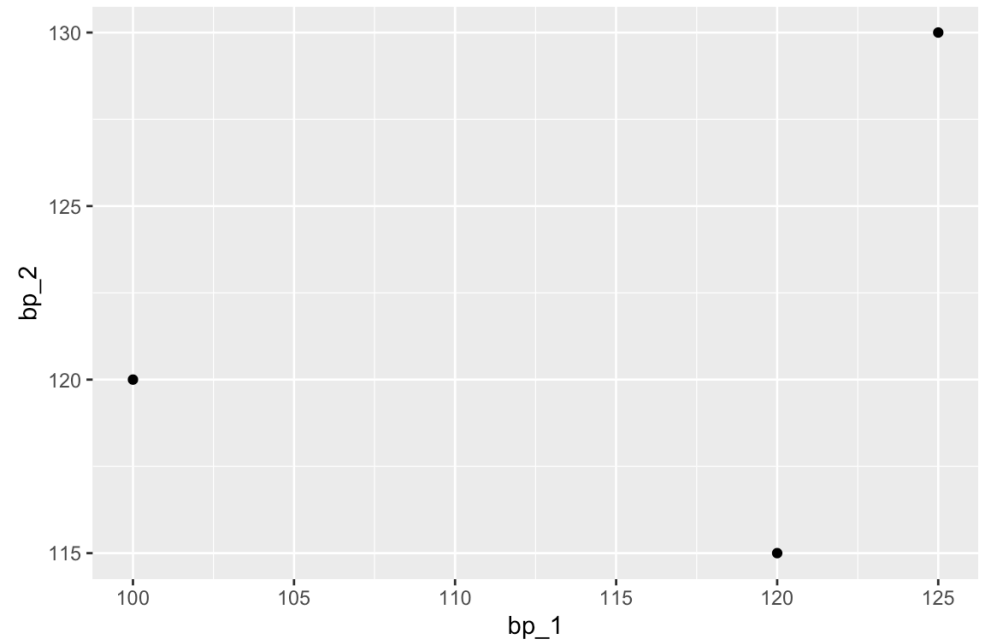
```
1 lm(literacy_rate ~ year, data = litF_long)
```


Another example from last time

```
1 df_3
```

	id	bp_1	bp_2	hr_1	hr_2
1	1	100	120	60	77
2	2	120	115	75	81
3	3	125	130	80	93

```
1 df_3 |>  
2   ggplot(aes(x = bp_1,  
3               y = bp_2)) +  
4   geom_point()
```

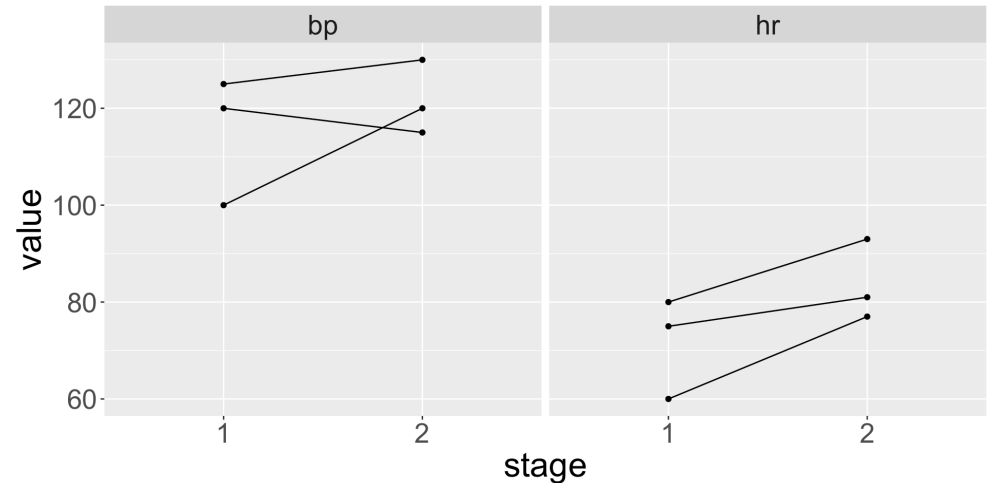


What we can do with reshaped data

```
1 df3_long <- df_3 |>
2   pivot_longer(cols = -id,
3                 names_to = c("measurement",
4                               "stage"),
5                 names_sep = "_",
6                 values_to = "value")
7
8 df3_long
```

A tibble: 12 × 4

	id	measurement	stage	value
	<dbl>	<chr>	<chr>	<dbl>
1	1	bp	1	100
2	1	bp	2	120
3	1	hr	1	60
4	1	hr	2	77
5	2	bp	1	120
6	2	bp	2	115
7	2	hr	1	75
8	2	hr	2	81
9	3	bp	1	125
10	3	bp	2	130
11	3	hr	1	80
12	3	hr	2	93



```
1 df3_long |>
2   ggplot(aes(x = stage,
3               y = value)) +
4   geom_point() +
5   geom_line(aes(group = id)) +
6   facet_wrap(~measurement)
```

Warmup activity

Work on the activity (handout) with a neighbor, then we will discuss as a class

Warmup

```
1 df_3
2
3 df_3 |>
4   pivot_longer(cols = -id,
5                 names_to = c(".value", "stage"),
6                 names_sep = "_")
```

	id	bp_1	bp_2	hr_1	hr_2
1	1	100	120	60	77
2	2	120	115	75	81
3	3	125	130	80	93

	id	stage	bp	hr
1	1	1	100	60
2	1	2	120	77
3	2	1	120	75
4	2	2	115	81
5	3	1	125	80
6	3	2	130	93

What is `names_to = c(".value", "stage")` doing?

What we can do with the reshaped data

```
# A tibble: 6 × 4
```

	id	stage	bp	hr
	<dbl>	<chr>	<dbl>	<dbl>
1	1	1	100	60
2	1	2	120	77
3	2	1	120	75
4	2	2	115	81
5	3	1	125	80
6	3	2	130	93

Call:

```
lm(formula = bp ~ hr + stage, data =  
df3_new)
```

Coefficients:

(Intercept)	hr	stage2
38.022	1.074	-6.223

```
1 lm(bp ~ hr + stage,  
2   data = df3_new)
```

Going the other way

Data on air quality in two locations (BETR801, London Westminster) on different days:

```
1 air_quality
```

```
# A tibble: 1,825 × 3
```

	date.utc	location	value
	<dtm>	<chr>	<dbl>
1	2019-06-18 06:00:00	BETR801	18
2	2019-06-17 08:00:00	BETR801	6.5
3	2019-06-17 07:00:00	BETR801	18.5
4	2019-06-17 06:00:00	BETR801	16
5	2019-06-17 05:00:00	BETR801	7.5
6	2019-06-17 04:00:00	BETR801	7.5
7	2019-06-17 03:00:00	BETR801	7
8	2019-06-17 02:00:00	BETR801	7
9	2019-06-17 01:00:00	BETR801	8
10	2019-06-16 01:00:00	BETR801	15

What if I want a separate column for each location?

pivot_wider

```
# A tibble: 3 × 3
```

	date.utc		location	value
	<dtm>		<chr>	<dbl>
1	2019-06-18 06:00:00		BETR801	18
2	2019-06-17 08:00:00		BETR801	6.5
3	2019-06-17 07:00:00		BETR801	18.5

```
1 air_quality |>
2   pivot_wider(id_cols = ...,
3               names_from = ...,
4               values_from = ...)
```

pivot_wider

```
# A tibble: 3 × 3
```

	date.utc	location	value
	<dtm>	<chr>	<dbl>
1	2019-06-18 06:00:00	BETR801	18
2	2019-06-17 08:00:00	BETR801	6.5
3	2019-06-17 07:00:00	BETR801	18.5

```
1 air_quality |>
2   pivot_wider(id_cols = date.utc,
3               names_from = location,
4               values_from = value)
```

```
# A tibble: 1,670 × 3
```

	date.utc	BETR801	`London Westminster`
	<dtm>	<dbl>	<dbl>
1	2019-06-18 06:00:00	18	7
2	2019-06-17 08:00:00	6.5	6
3	2019-06-17 07:00:00	18.5	6
4	2019-06-17 06:00:00	16	6
5	2019-06-17 05:00:00	7.5	6
6	2019-06-17 04:00:00	7.5	6
7	2019-06-17 03:00:00	7	6

8	2019-06-17	02:00:00	7	6
9	2019-06-17	01:00:00	8	6
10	2019-06-16	01:00:00	15	7

Class activity

- Work with a neighbor on the class activity
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)

For next time, read:

- Chapter 5 in *Modern Data Science with R*